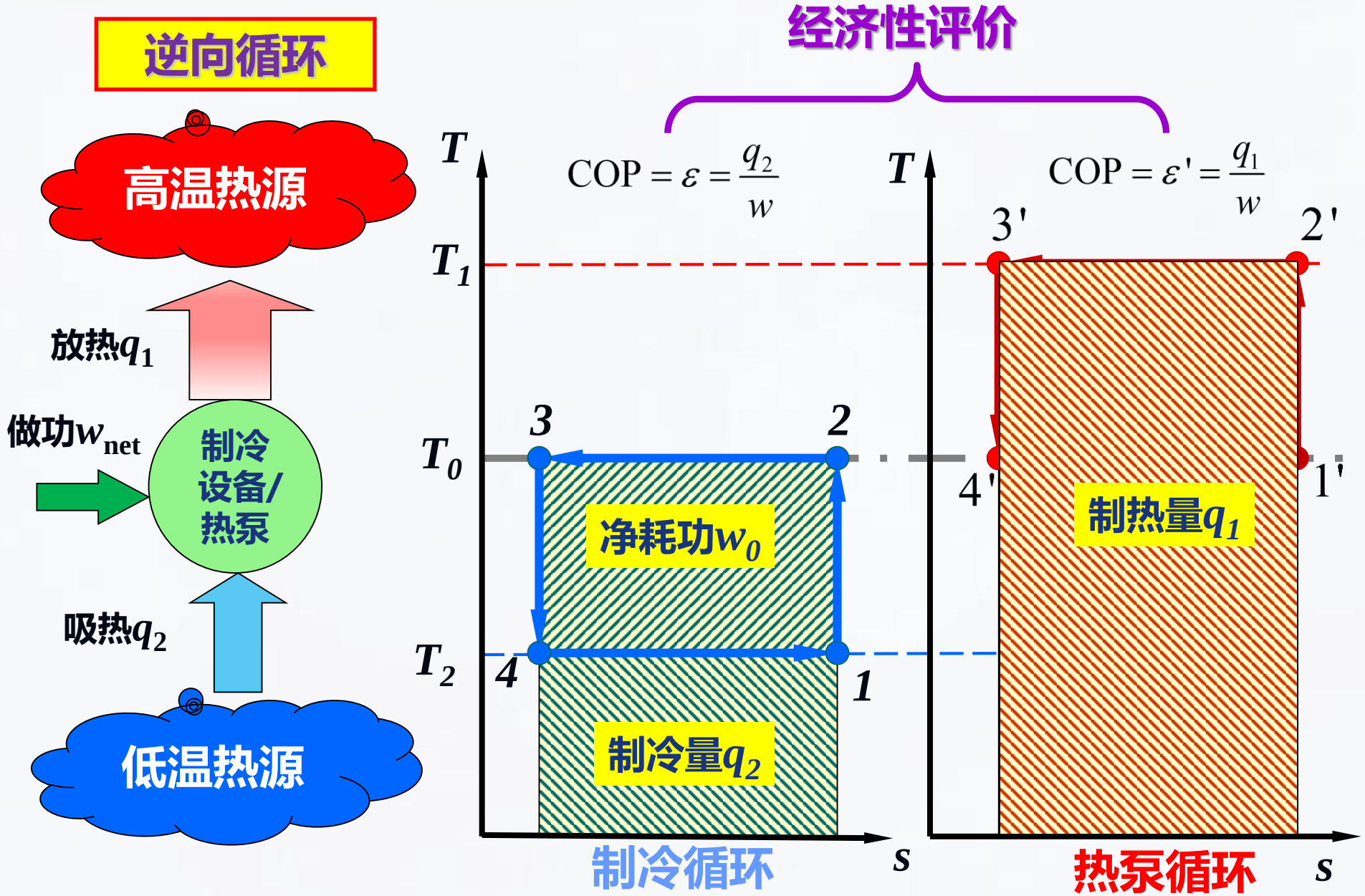
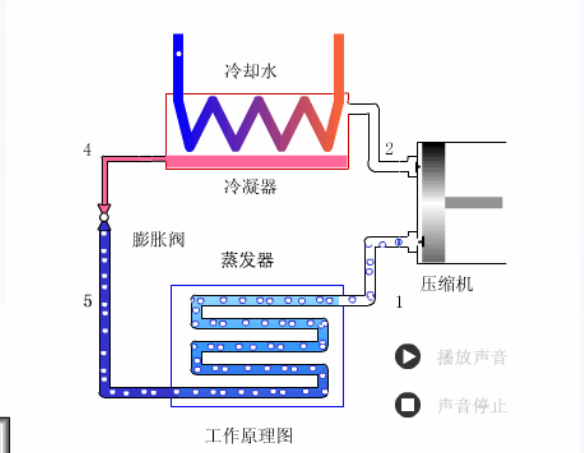
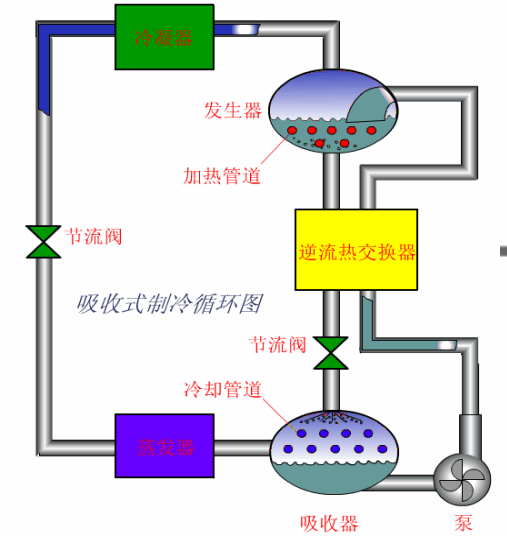

逆向循环
高温热源
放热q_1
做功w_{net}
制冷设备/热泵
吸热q_2
低温热源
经济性评价
T
T_1
T_0
T_2
$\mathrm{COP}=\varepsilon=\frac{q_2}{w}$
3
2
4
1
净耗功w_0
制冷量q_2
s
制冷循环
T
$\mathrm{COP}=\varepsilon'=\frac{q_1}{w}$
3'
2'
4'
1'
制热量q_1
s
热泵循环

制冷循环类型

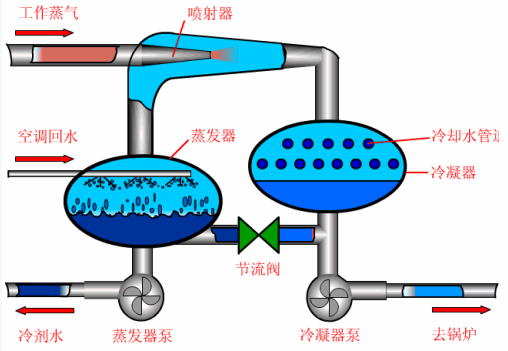
- 压缩式制冷循环
  - 压缩气体制冷
  - 压缩蒸气制冷
- 吸收式制冷循环
- 吸附式制冷循环
- 蒸气喷射制冷循环
- 半导体制冷
- 热声制冷
- 磁制冷



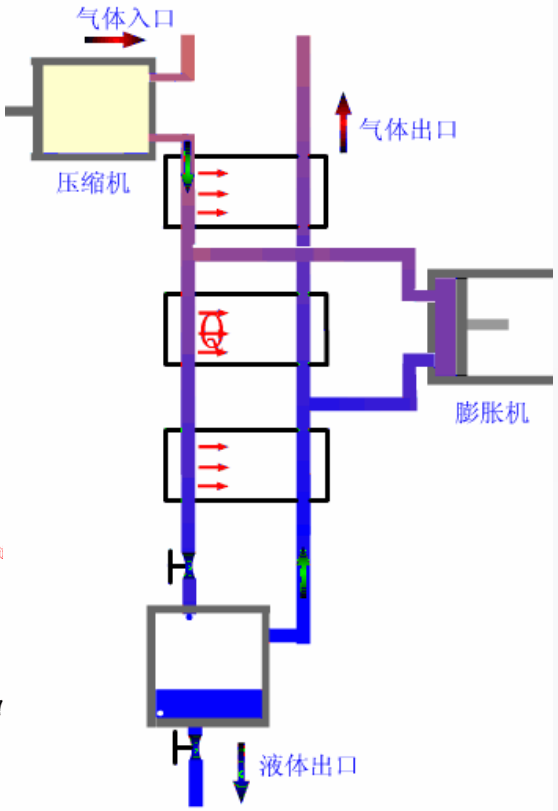
工作原理图

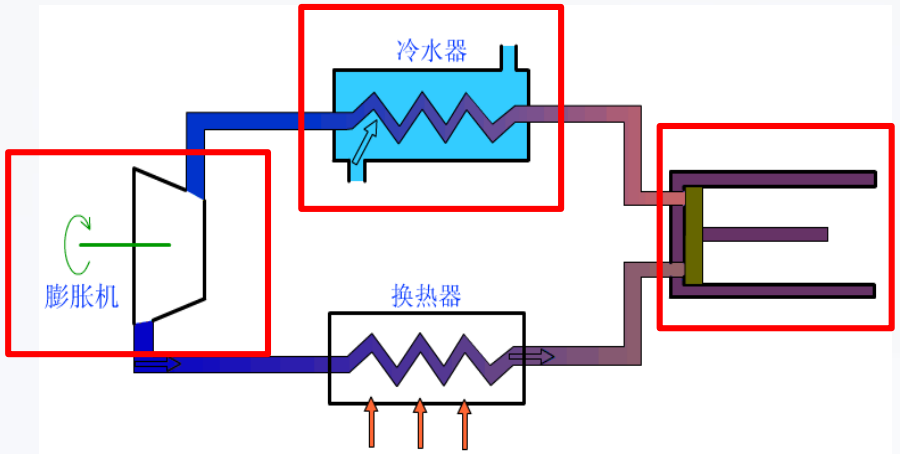


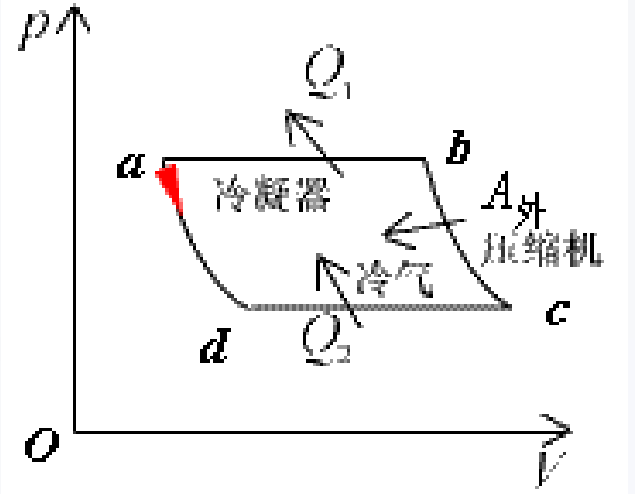
吸收式制冷循环图



蒸气喷射式制冷系统







四个主要部件；工质：**空气**

理想化处理：①理气； ②定比热； ③可逆；

c⟶**b** 绝热压缩 p↑ T↑；　　**b**⟶**a** 等压冷却 向环境放热，T↓

a⟶**d** 绝热膨胀 T↓ $<T_1$（**冷库**）；　**d**⟶**c** 等压吸热 T↑⟶T_1

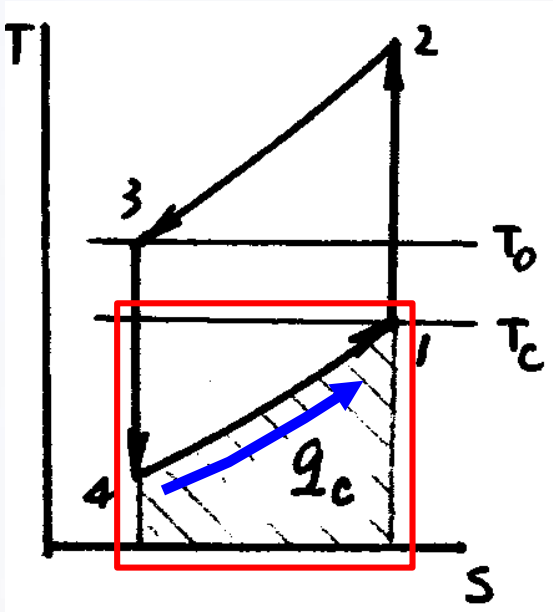
制冷系数 Coefficient of performance(COP)

$$\varepsilon = \frac{q_c}{w_{net}} = \frac{q_c}{q_1 - q_c} \qquad q_1 = h_2 - h_3 \qquad q_c = h_1 - h_4$$

$$\varepsilon = \frac{q_c}{w_{net}} = \frac{h_1 - h_4}{(h_2 - h_3) - (h_1 - h_4)} \overset{?}{=} \frac{T_1 - T_4}{(T_2 - T_3) - (T_1 - T_4)}$$

$\because$ $T_2 = T_1 \pi^{\frac{\kappa-1}{\kappa}}$, $T_3 = T_4 \pi^{\frac{\kappa-1}{\kappa}}$

$\therefore$ $= \dfrac{1}{\pi^{\frac{\kappa-1}{\kappa}} - 1} = \dfrac{T_1}{T_2 - T_1}$



讨论： 1）相同温度的T_0和T_C ➡ $\varepsilon_c = \dfrac{T_c}{T_0 - T_c} > \dfrac{T_1}{T_2 - T_1} = \varepsilon$

2）$\pi\uparrow$ $\varepsilon\downarrow$, 但$q_c\uparrow$

空气压缩制冷循环特点：

- **优点**：工质无毒，无味，不怕泄漏。
- **缺点**：

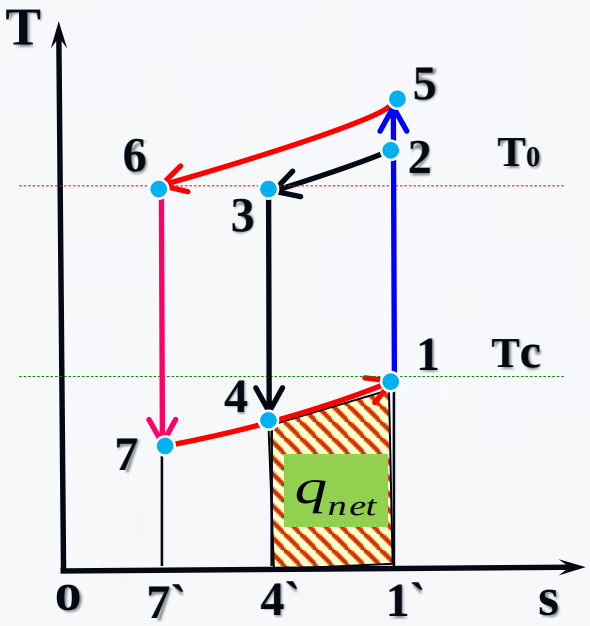
1.无法实现 T ，$\varepsilon < \varepsilon_C$

2. $q_2 = c_p(T_1 - T_4)$，空气c_p很小，q_2 很小。

若$(T_1 - T_4)$ ↑ ε ↓

3. 活塞式流量m小，制冷量$Q_2 = m\ q_2$小。

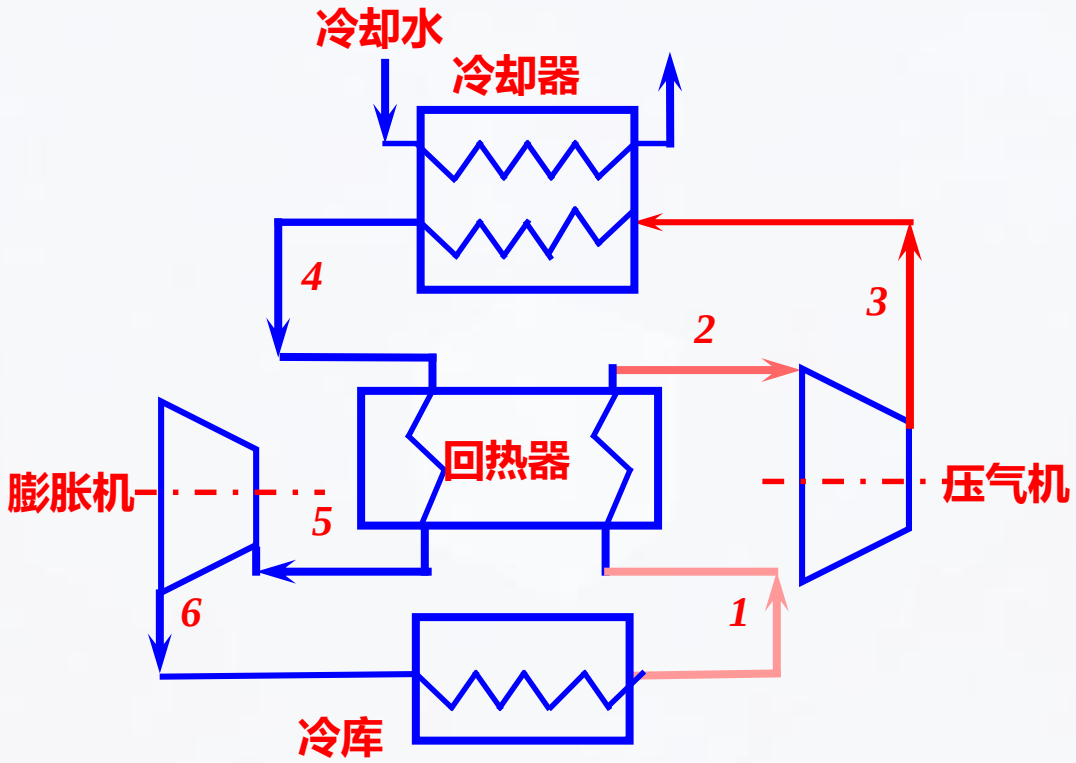
使用叶轮式，再回热则可用。



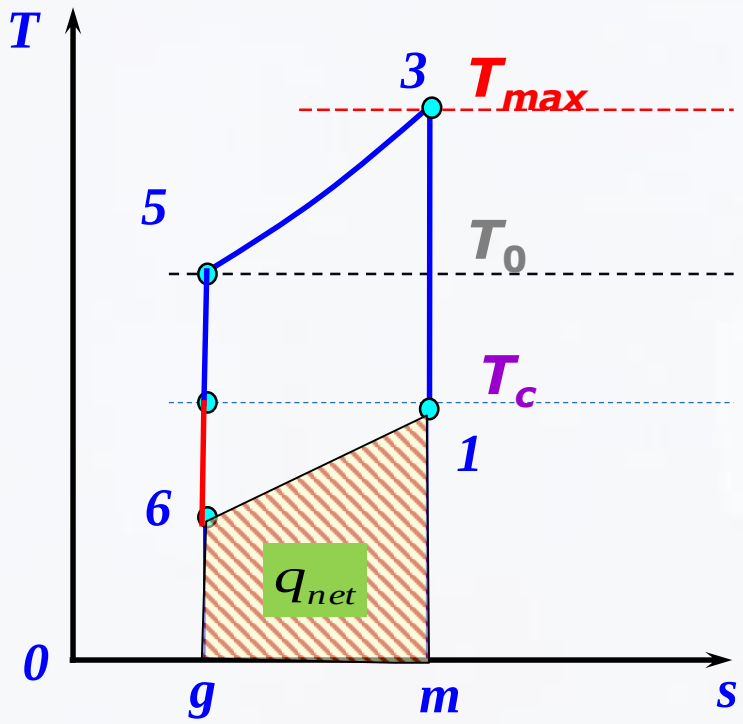
回热式压缩空气制冷循环

压缩空气制冷，q_c较小，且随π上升，ε下降，为兼顾 Q_c及ε，采用大流量叶轮压缩机并回热。

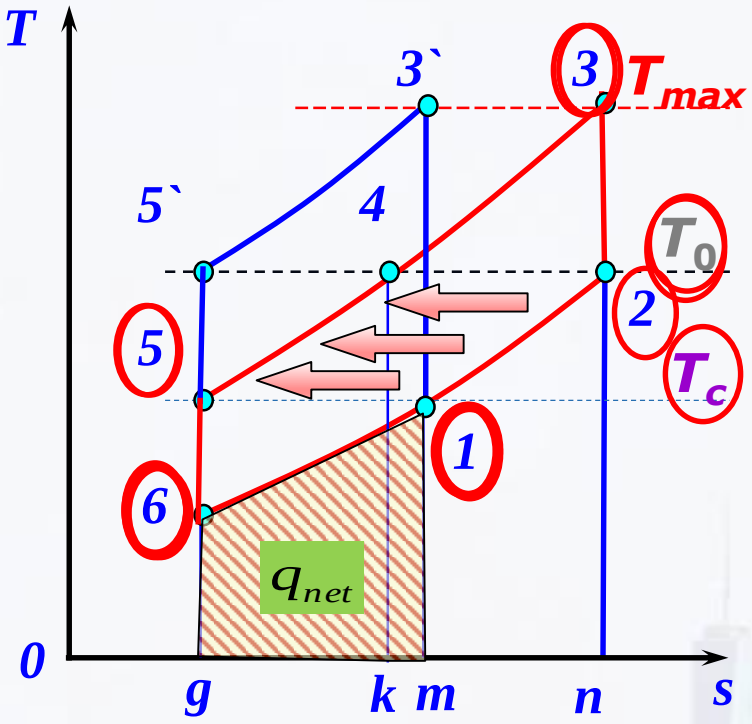




回热式压缩空气制冷系统



简单压缩空气制冷循环T-s图

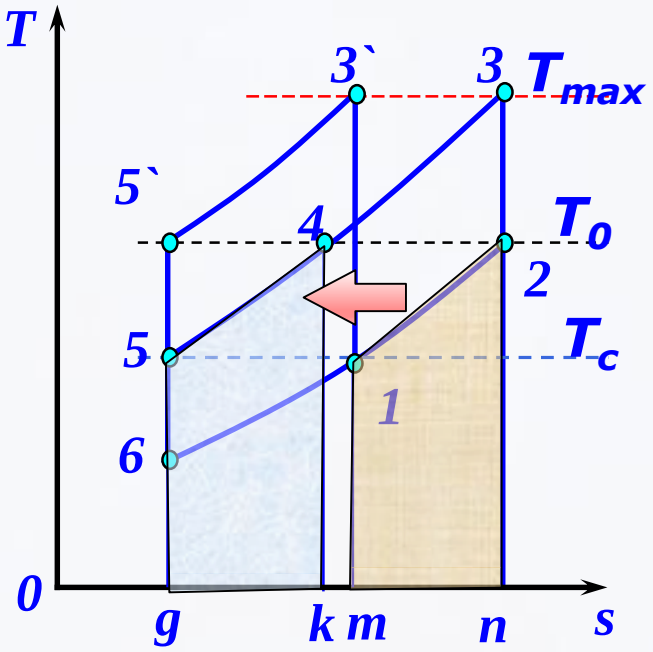


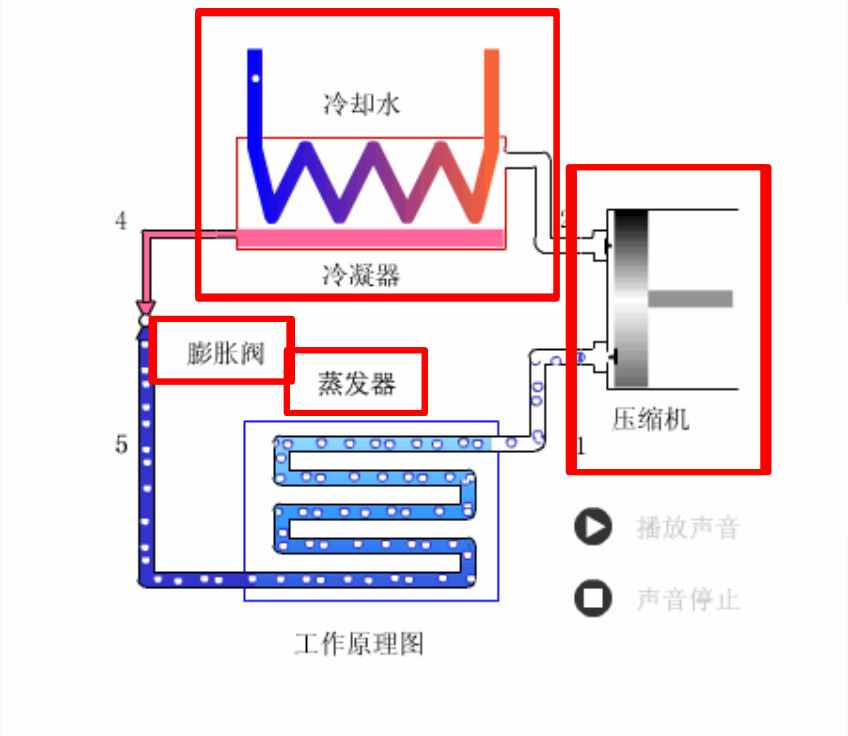
回热式压缩空气制冷循环T-s图

回热后： ε相等， π下降

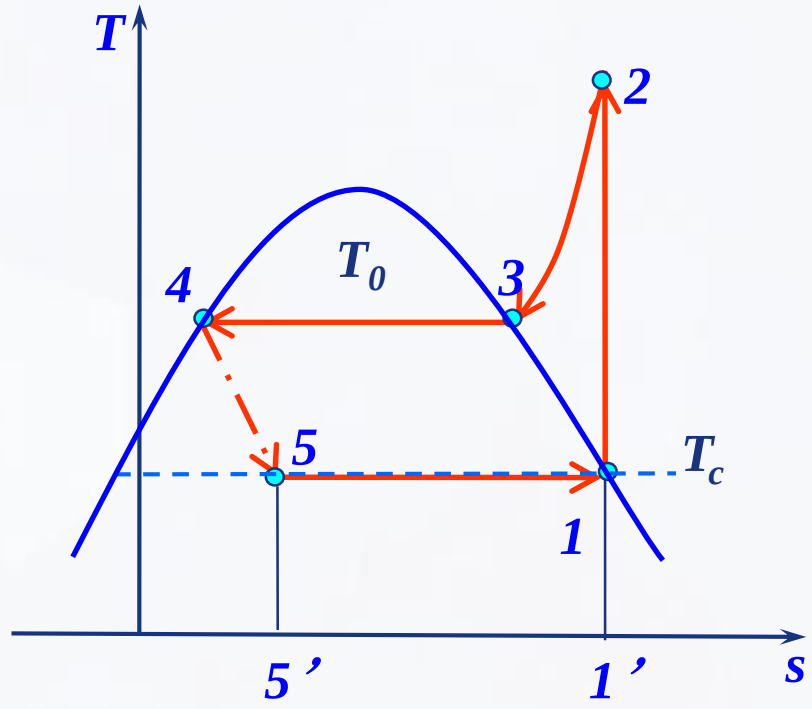
回热循环优点

- 同样制冷系数下，增压比下降，这为采用小压比大流量的叶轮式压气机和膨胀机提供可能；
- 增压比减小，使压缩过程和膨胀过程的不可逆损失减小。





工作原理图



压缩蒸气制冷循环*T-s*图

蒸发器中吸热量:

$$q_2 = h_1 - h_5 = h_1 - h_4$$

冷凝器中放热量: $q_1 = h_2 - h_4$

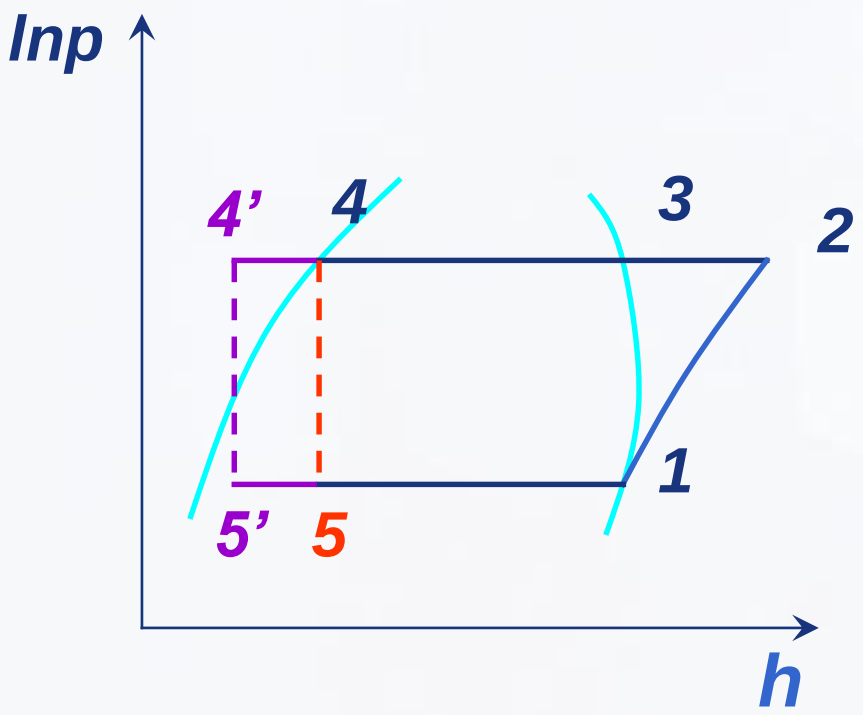
制冷系数:

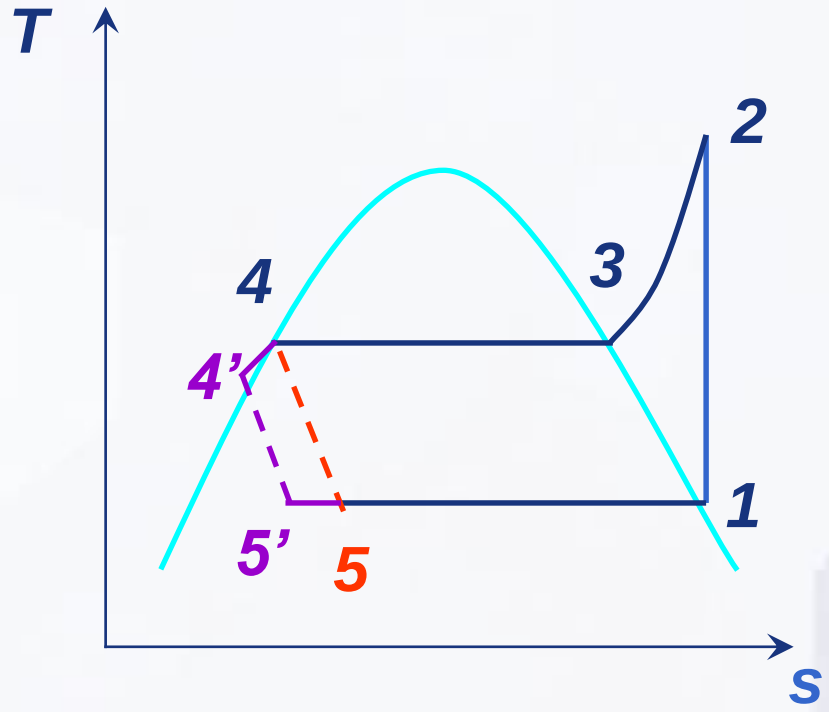
$$\varepsilon = \frac{q_2}{q_1 - q_2} = \frac{h_1 - h_4}{(h_2 - h_4) - (h_1 - h_4)} = \frac{h_1 - h_4}{h_2 - h_1} = \frac{q_2}{w}$$

$$\times = \frac{T_1 - T_4}{T_2 - T_1}$$



工程上常用过冷措施

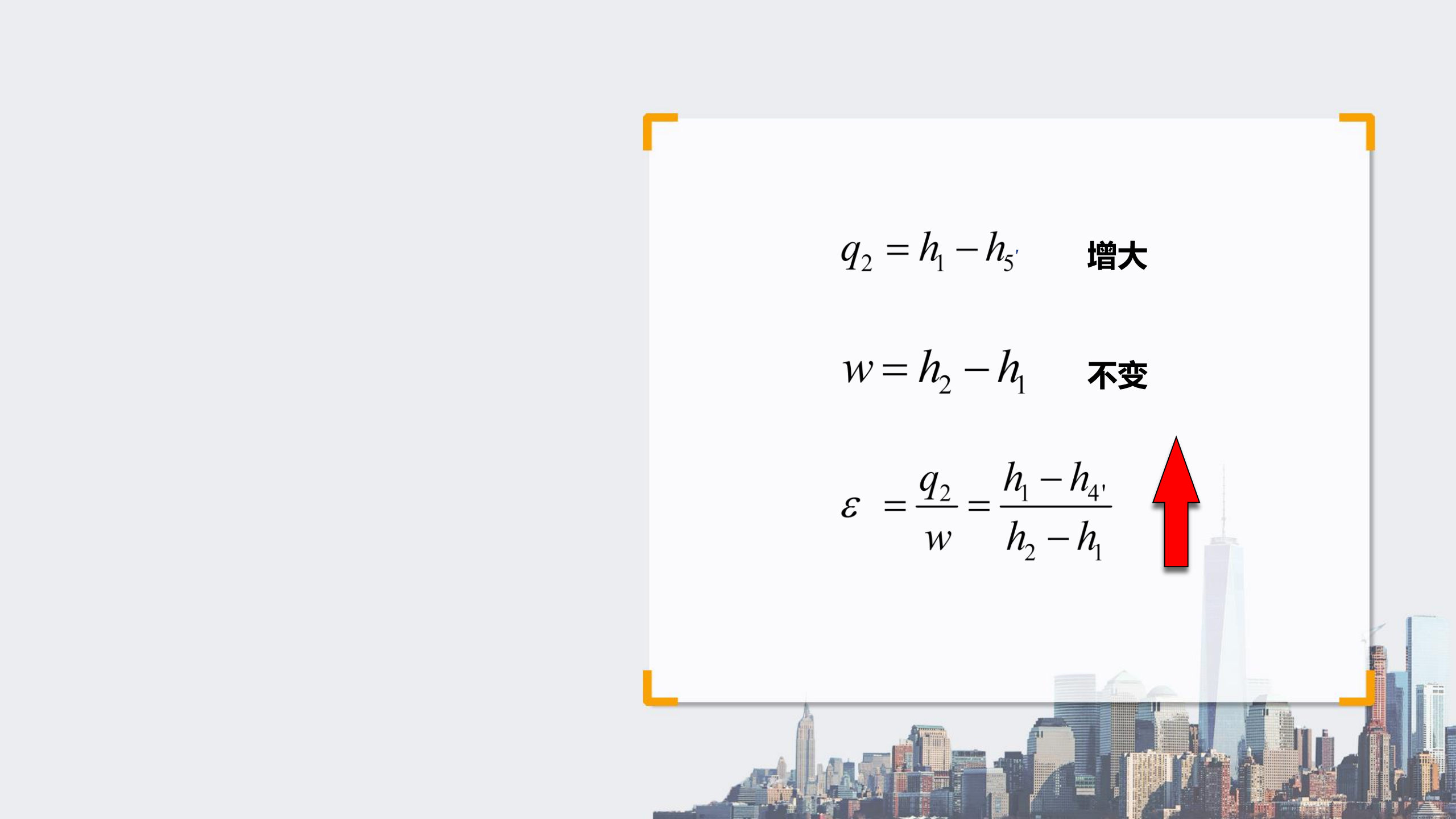




$$q_2 = h_1 - h_{5'}$$ **增大**

$$w = h_2 - h_1$$ **不变**

$$\varepsilon = \frac{q_2}{w} = \frac{h_1 - h_{4'}}{h_2 - h_1}$$



逆卡诺热机及膨胀机替代节流阀的分析

优点：

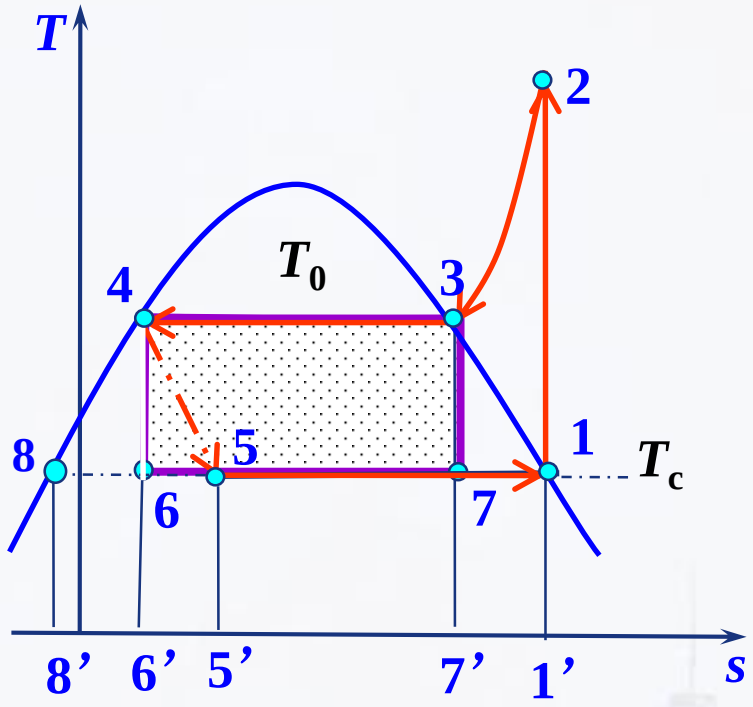
1. 回收功量　　$h_4 - h_6$

2. 制冷量有所增加?

$$h_5 - h_6 = h_4 - h_6$$　　**面积8468**

$$h_4 - h_8 - (h_6 - h_8)$$

面积8'8466'　面积8'866'



缺点：1.设备复杂，两相压缩或膨胀存在安全隐患；

2. 7-1过程的制冷能力没有得到发挥；

弊>利